

T.R.
GEBZE TECHNICAL UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

**MMWAVE RADAR-BASED VOWEL
RECOGNITION FOR GAMIFIED SPEECH
THERAPY**

SELIN DURU DERINDAG

**SUPERVISOR
PROF. DR. YUSUF SINAN AKGUL**

**GEBZE
2025**

T.R.
GEBZE TECHNICAL UNIVERSITY
FACULTY OF ENGINEERING
COMPUTER ENGINEERING DEPARTMENT

**MMWAVE RADAR-BASED VOWEL
RECOGNITION FOR GAMIFIED SPEECH
THERAPY**

SELIN DURU DERINDAG

SUPERVISOR
PROF. DR. YUSUF SINAN AKGUL

2025
GEBZE

 GEBZE TECHNICAL UNIVERSITY	GRADUATION PROJECT JURY APPROVAL FORM
--	--

This study has been accepted as an Undergraduate Graduation Project in the Department of Computer Engineering on 26/06/2025 by the following jury.

JURY

Member

(Supervisor) : Prof. Dr. Yusuf Sinan Akgul

Member : Assoc. Prof. Dr. Mehmet GÖKTÜRK

ABSTRACT

This study presents a novel speech therapy system that uses millimeter-wave (mmWave) radar and convolutional neural networks (CNNs) to classify Turkish vowel articulations without relying on audio input. Designed for individuals with hearing impairments, particularly those who are profoundly deaf, the system captures fine facial and throat movements via radar and converts them into spectrograms for image-based classification. A real-time feedback game visually maps classified vowels onto an articulatory chart, allowing users to adjust their pronunciation based on dynamic feedback. The proposed method achieved a test accuracy of 49.1% on unseen radar spectrogram data, demonstrating the viability of non-acoustic vowel recognition. Despite challenges in distinguishing visually similar vowels such as /o/ and /ü/, the system proved robust in detecting articulatory patterns and responsive in live gameplay scenarios. This work contributes a privacy-preserving, microphone-free approach to speech therapy and lays the groundwork for future multimodal rehabilitation tools.

Keywords: mmWave radar, speech therapy, spectrogram image classification, radar speech recognition, non-acoustic speech interface, real-time signal processing, convolutional neural networks (CNN), visual game-based feedback.

ÖZET

Bu çalışma, işitme engelli bireyler için sesli girdiye ihtiyaç duymadan Türkçe ünlü harflerin sınıflandırılmasını sağlayan yeni bir konuşma terapisi sistemi sunmaktadır. Sistem, milimetre dalga (mmWave) radar teknolojisi ve evrişimli sinir ağları (CNN) kullanarak yüz ve boğaz bölgesindeki mikro hareketleri algılar ve bu verileri spektrogramlara dönüştürerek görüntü tabanlı sınıflandırma gerçekleştirir. Gerçek zamanlı geri bildirim sağlayan bir oyun arayüzü, sınıflandırılan ünlüleri artikülasyon tablosu üzerinde görselleştirerek kullanıcının telaffuzunu dinamik olarak düzeltmesini sağlar. Önerilen yöntem, görülmemiş radar spektrogram verilerinde %49,1 test doğruluğu elde etmiş ve akustik olmayan ünlü tanınmanın uygulanabilirliğini göstermiştir. /o/ ve /ü/ gibi görsel olarak benzer ünlülerin ayırt edilmesinde zorluklar yaşansa da, sistem artikülasyon desenlerini güvenilir biçimde tanıyabilmiş ve canlı kullanım senaryolarında akıcı geri bildirim sağlamıştır. Bu çalışma, mahremiyeti koruyan ve mikrofona ihtiyaç duymayan bir konuşma terapisi yaklaşımı sunarak gelecekteki çok modlu rehabilitasyon araçları için temel oluşturmaktadır.

Anahtar Kelimeler: mmWave radar, konuşma terapisi, spektrogram görüntü sınıflandırması, radarla konuşma tanıma, akustik olmayan konuşma arayüzü, gerçek zamanlı sinyal işleme, convolutional neural networks (CNN), görsel ve oyun tabanlı geri bildirim.

ACKNOWLEDGEMENT

I'd like to acknowledge my parents, Banu and Mahmut Derindag to whom I owe an immense debt of gratitude for their unconditional support and unwavering patience throughout my undergraduate pursuits, without which I would not have made it this far.

Salih Arda Kizil, for his endless patience, understanding, and guidance, especially during the most difficult times I faced during my studies.

My academic supervisor, Prof. Dr. Yusuf Sinan Akgul, for guiding me since the beginning of my undergraduate studies, and Res. Asst. İlhan Aytutuldu for helping throughout my graduation project.

I am also thankful for my friends, who provided both distractions when needed and encouragement when it seemed impossible to continue. The company of my peers have enriched my learning experience, making this journey not only possible but also enjoyable.

Selin Duru Derindag

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol or

Abbreviation : Explanation

CNN : Convolutional Neural Network

STFT : Short-time Fourier transform

HOH : Hard of hearing

FFT : Fast Fourier Transform

CONTENTS

Abstract	iv
Özet	v
Acknowledgement	vi
List of Symbols and Abbreviations	vii
Contents	x
List of Figures	xi
List of Tables	xii
1 Introduction	1
2 Background and Related Work	2
2.1 Speech Therapy and Visual Feedback Systems	2
2.1.1 The IPA Vowel Chart and Articulatory Features	3
2.2 mmWave Radar for Speech Detection	4
2.3 Image Classification	4
2.4 Computer Game-Based Rehabilitation	5
3 System Design and Methodology	6
3.1 Hardware Configuration	6
3.1.1 mmWave Radar Sensor	6
3.1.2 DCA1000EVM Data Capture Card	6
3.1.3 Host Computer	6
3.1.4 Peripherals	7
3.2 Data Acquisition	7
3.3 Pre-processing: From ADC to Spectrogram	7
3.3.1 Loading and Padding	7
3.3.2 Reshaping	8
3.3.3 Channel Averaging	8
3.3.4 STFT Transformation	8
3.3.5 Spectrogram Image Generation	8

3.4	CNN Architecture and Training Strategy	9
3.4.1	Model Architecture	9
3.4.2	Training Strategy	9
3.4.2.1	Experiment 1: Enhanced CNN Architecture with Moderate Class Weighting	10
3.4.2.2	Experiment 2: Enhanced CNN with Stronger Weight- ing Strategy	10
3.4.2.3	Experiment 3: Alternate Architecture with Entropy Regularization	11
3.5	Real-Time Inference Pipeline	11
3.5.1	Radar Data Acquisition and Processing	11
3.5.2	Spectrogram Display and Interface	12
3.5.3	CNN-Based Classification	12
3.6	Game-Based Visualization and Classifier Integration	12
3.6.1	Game design and Layout	12
3.6.2	Real Time Prediction Integration	13
4	Experiment Results	14
4.1	Classification Accuracy	14
4.1.1	Confusion Matrix Patterns Among Experimented Architectures	14
4.1.1.1	Experiment 1 - Enhanced Architecture (Best Overall Performance	14
4.1.1.2	Experiment 2- Extreme Class Weighting	15
4.1.1.3	Experiment 3 - Conservative Model	15
4.1.2	Per-Class Analysis for Experiment 1	16
4.2	Real Time Performance Metrics	18
4.2.1	Frame Rate and Inference Frequency	18
4.2.2	Inference Latency	18
4.2.3	Observed Inference Behavior and Bias	18
4.2.4	Gameplay Responsiveness	19
5	Discussion	20
5.1	Interpretation of Results	20
5.2	Future Work	21
5.2.1	Dual-Radar Configuration	21
5.2.2	Expanded Dataset and Speaker Diversity	21
5.2.3	Integration with Clinical Protocols	21
6	Conclusion	22

LIST OF FIGURES

2.1	IPA Vowel Placement Chart.	3
3.1	Spectrogram image generated from radar data.	8
4.1	Confusion matrix result of experiment 1.	15
4.2	Confusion matrix result of experiment 2.	16
4.3	Confusion matrix result of experiment 3.	16

LIST OF TABLES

4.1	Per-class performance of the CNN model on the radar spectrogram test set (Experiment 1)	17
-----	---	----

1. INTRODUCTION

Effective speech therapy often relies on auditory feedback to guide individuals in adjusting their pronunciation. However, for people who are hard of hearing or profoundly deaf, traditional auditory-based methods may be ineffective or inaccessible. In such cases, alternative modalities—particularly visual and tactile feedback—become essential. While lip-reading and visual cues from speech therapists can aid articulation, these approaches offer limited access to internal articulatory dynamics like tongue and glottal positioning, which are crucial for accurate vowel production. Recent advances in radar sensing have enabled new possibilities for non-contact, privacy-preserving observation of speech-related movements. Millimeter-wave (mmWave) radar, in particular, can detect micro-movements in facial and throat musculature at sub-millimeter precision, offering a promising alternative to traditional audio-based speech analysis. When combined with modern deep learning techniques such as convolutional neural networks (CNNs), radar signals can be transformed into informative visual representations, such as spectrograms, suitable for classification tasks. This thesis proposes a real-time vowel classification and feedback system that uses mmWave radar to detect articulatory gestures associated with five Turkish vowels: /a/, /e/, /i/, /o/, and /ü/. The system generates spectrograms from radar data and applies a CNN to identify the spoken vowel, which is then displayed on a visual chart through a game interface. This approach enables users to receive immediate, intuitive feedback about their pronunciation, thereby supporting the development of accurate speech patterns without the need for microphones or audio processing. The remainder of this thesis details the system architecture, data collection and pre-processing pipeline, model design, and real-time feedback implementation. Evaluation results highlight both the promise and current limitations of radar-based vowel recognition, pointing to future improvements in model generalization, user adaptation, and clinical applicability.

2. BACKGROUND AND RELATED WORK

2.1. Speech Therapy and Visual Feedback Systems

Hearing plays a fundamental role in speech development; any impairment in hearing can significantly disrupt one's ability to produce clear speech. Individuals who are born hard of hearing (HOH) or experience hearing loss during early developmental stages often struggle to perceive how specific sounds and vowels are articulated. This is largely due to their inability to audibly or visually discern internal articulatory movements such as tongue positioning and glottal activity. Vowels that share similar lip positions can be especially challenging to differentiate. Children with hearing loss often miss out on perceiving particular sound frequencies, rendering them unable to replicate certain consonants or vowels. Consequently, their challenges extend beyond hearing impairment to profound difficulties in speech and language development[1]. Although lip-reading skills can be cultivated from a young age, a child cannot reproduce a sound they cannot hear. In fully deaf children, speech production relies solely on visual cues, which are often insufficient and result in inaccurate pronunciation. Auditory-verbal therapy (AVT) is one established method for addressing speech development in hearing-impaired children. This early intervention approach uses technological aids like hearing devices or cochlear implants to enable listening and speaking skills[2]. However, such devices can be expensive, uncomfortable, or ineffective for children with profound hearing loss. In these cases, speech therapy shifts focus toward articulation training, emphasizing visible features like tongue placement, mouth shape, and airflow. Success in this method depends on repetitive and intensive practice to build articulatory muscle memory[3]. While visual input and lip reading assist in this process, they are not sufficient alone. For fully deaf individuals, visual feedback systems, like the one proposed in this thesis, offer a promising alternative to auditory tools. Several visual biofeedback tools have been introduced to aid articulation. Opti-Speech, developed by Katz et al. (2014)[4], is a notable example. It employs electromagnetic articulography to track real-time tongue and jaw movements and presents them inside a 3D virtual head model. Although effective in adults, the system may appear unintuitive or monotonous for children and can be uncomfortable due to the need for physical sensors. Still, it exemplifies how real-time visual feedback can enhance speech accuracy in therapeutic contexts.

2.1.1. The IPA Vowel Chart and Articulatory Features

The International Phonetic Alphabet (IPA) offers a standardized method for describing speech sounds, particularly vowels. The IPA vowel chart is arranged as a trapezoid that maps tongue position (height and backness) and lip shape (rounded vs. unrounded). In Turkish, the target vowels in this study — /a/, /e/, /i/, /o/, and /ü/ — are distributed across the IPA chart as follows:

- /i/: high front unrounded
- /e/: mid front unrounded
- /a/: low central unrounded
- /o/: mid back rounded
- /ü/: high front rounded

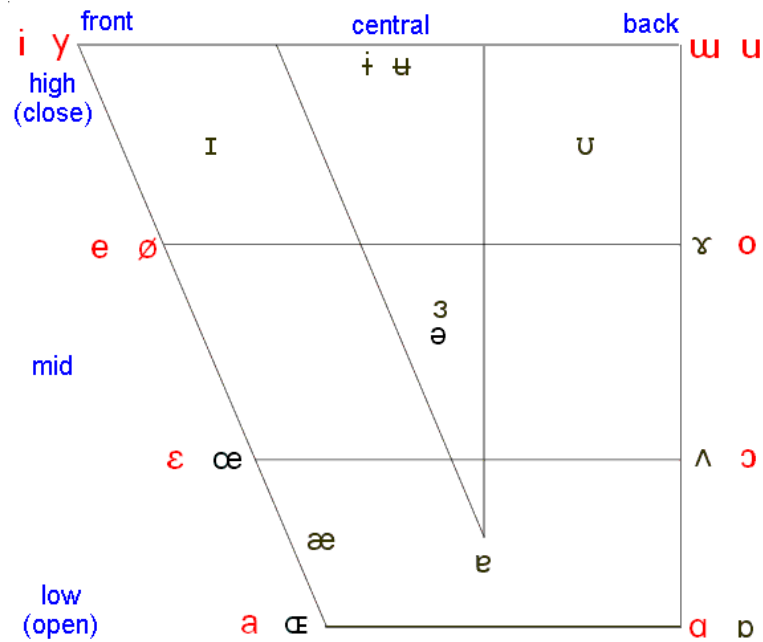


Figure 2.1: IPA Vowel Placement Chart.

Certain vowels (e.g., /o/ and /ü/) appear similar in lip movement but differ internally in tongue positioning. These subtle differences are difficult to perceive visually, particularly for those relying entirely on visual cues. By aligning the game's visual design with the IPA trapezoid, this project reinforces a linguistically sound framework that directly connects feedback visuals with articulatory mechanics.

2.2. mmWave Radar for Speech Detection

Millimeter-wave (mmWave) radar is a non-contact, privacy-preserving sensing technology capable of detecting fine human movements, including speech-related articulatory gestures. Operating typically between 76–81 GHz, mmWave radar transmits high-frequency electromagnetic waves that reflect off nearby objects. Analyzing these return signals provides precise measurements of distance, motion, and angle. Due to its short wavelength (around 4mm), mmWave radar can detect micro-movements at sub-millimeter levels, making it suitable for capturing articulatory motions such as lip and throat vibrations[5]. Its non-intrusive nature makes it ideal for therapeutic settings. Recent studies demonstrate the feasibility of radar-based speech detection. Steinmetz and Balal (2025) used a 94 GHz micro-Doppler radar to reconstruct speech signals from vocal cord vibrations, achieving strong correlation with ground-truth audio using STFT-based reconstruction methods[6]. Similarly, Fan et al. (2023) introduced mmMIC, a single-radar system capable of extracting both lip and glottal features using a transformer-based fusion model, reaching 92% accuracy without microphones [7]. Lenz et al. (2025) further developed a biomechanical model that links radar-measured neck vibrations with acoustic speech signals, supporting the theoretical underpinnings of radar-based speech recognition[8].

2.3. Image Classification

To enable real-time classification, this system converts radar signals into spectrograms and uses image classification models to recognize vowel patterns. Image classification assigns a label to an entire image, unlike object detection, which localizes multiple items within an image[9]. Convolutional Neural Networks (CNNs) are especially suited for image classification due to their ability to extract hierarchical features. They use convolutional layers for pattern detection, pooling layers for dimensionality reduction, and dense layers for final classification. CNNs like LeNet-5, AlexNet, and ResNet have demonstrated success in various fields including health-care and autonomous driving [10]. Audio classification via CNNs involves converting 1D audio into 2D spectrograms, allowing models to analyze time-frequency patterns. This approach has shown effectiveness in tasks like speech command recognition and environmental sound detection[11].

2.4. Computer Game-Based Rehabilitation

Digital games have proven beneficial in speech therapy, especially among children. Saeedi et al. (2022) reviewed 27 studies and found that game-based interventions improved engagement, motivation, and adherence to therapy. These games often targeted phoneme articulation using tools like MFCC and PocketSphinx. However, recognition errors and environmental noise were common challenges[12]. Ahmed et al. (2018) tested five speech-controlled mobile games and reported high engagement from children. Real-time ASR provided instant feedback, and features like difficulty levels and rewards enhanced motivation. Speech-language pathologists also supported the games' potential for home-based practiceFifteen. Furlong et al. (2017) reviewed 14 CBST (computer-based speech therapy) programs and concluded that such systems can yield measurable speech improvements, especially when incorporating visual feedback and gamification. They advocated for further clinical validation and stronger collaboration between developers and therapists to maximize therapeutic value[13]. This literature supports the integration of real-time game-based feedback in modern speech therapy, as implemented in the present thesis.

3. SYSTEM DESIGN AND METHODOLOGY

This section outlines the system components used for real-time acquisition, processing, classification, and feedback of radar-based vowel signals. The architecture integrates mmWave sensing hardware, custom data pipelines, and a game-based interface.

3.1. Hardware Configuration

To support real-time non-acoustic speech recognition, the system uses a mmWave radar sensor, a data capture card, and a host computer for inference and visualization.

3.1.1. mmWave Radar Sensor

The primary sensing device is the Texas Instruments IWR1843 radar, which operates in the 76–81 GHz frequency range. It includes an onboard DSP and MCU for initial signal preprocessing. Configured via UART using a custom script, the radar captures micro-movements in the face and throat. Key settings include:

- 128 chirps per frame
- 128 ADC samples per chirp
- Sampling rate: 10 MSPS

3.1.2. DCA1000EVM Data Capture Card

The DCA1000EVM interfaces with the IWR1843 via SPI and streams raw ADC data to the host computer over Ethernet. It is configured through mmWave Studio for initial setup; real-time operation is handled by a custom Python UDP listener.

3.1.3. Host Computer

Radar data is received, processed, and classified on a standard laptop (Intel Core i7, 12 GB RAM, Windows 10). Model training was conducted using Google Colab with GPU acceleration, but real-time inference runs on the CPU.

3.1.4. Peripherals

A standard monitor displays real-time spectrograms and game feedback. No microphones or audio devices are used, reinforcing the system’s non-acoustic design.

3.2. Data Acquisition

The dataset used in this study was obtained from a pre-existing radar speech corpus. The data acquisition process involved a group of up to 35 participants, each instructed to pronounce the five target vowels — /a/, /e/, /i/, /o/, and /ü/ — in isolation. Each vowel was sustained for approximately two seconds, and repeated at least 20 times per participant, resulting in a substantial and balanced dataset across vowel classes and speakers. The recordings were made using the Texas Instruments IWR1843 mmWave radar positioned in front of the subject at a fixed distance, with no use of microphones or other acoustic sensors. The radar captured micro-movements in facial muscles, throat, and surrounding soft tissue associated with vowel articulation. These signals were sampled at a high rate using the DCA1000EVM data capture card, generating raw ADC data for each utterance. Although the recordings were performed prior to the implementation phase of this study, the data aligns with the system’s intended application — classifying non-acoustic vowel signals using spectrogram-based image classification. The dataset was subsequently organized into per-vowel directories and used as the training and testing foundation for the CNN model developed in this work.

3.3. Pre-processing: From ADC to Spectrogram

The conversion of raw radar data into spectrograms involved a multi-step Python pipeline:

3.3.1. Loading and Padding

Binary files were read into NumPy arrays. If data was incomplete, near-zero padding was applied to match the expected shape: 10 frames $\times$ 4 receivers $\times$ 128 chirps $\times$ 128 samples

3.3.2. Reshaping

Data was restructured using custom helpers to ensure correct interpretation of radar returns.

3.3.3. Channel Averaging

Signals from four receivers were averaged to reduce dimensionality and noise.

3.3.4. STFT Transformation

A Short-Time Fourier Transform (STFT) was applied with parameters:

- Sampling rate: 640 Hz
- FFT size: 256
- Hop length: 128

The result was log-scaled using `librosa.amplitude_to_db`.

3.3.5. Spectrogram Image Generation

Spectrograms were saved as grayscale images (128×128), with no axes or labels, and sorted into vowel-specific folders.

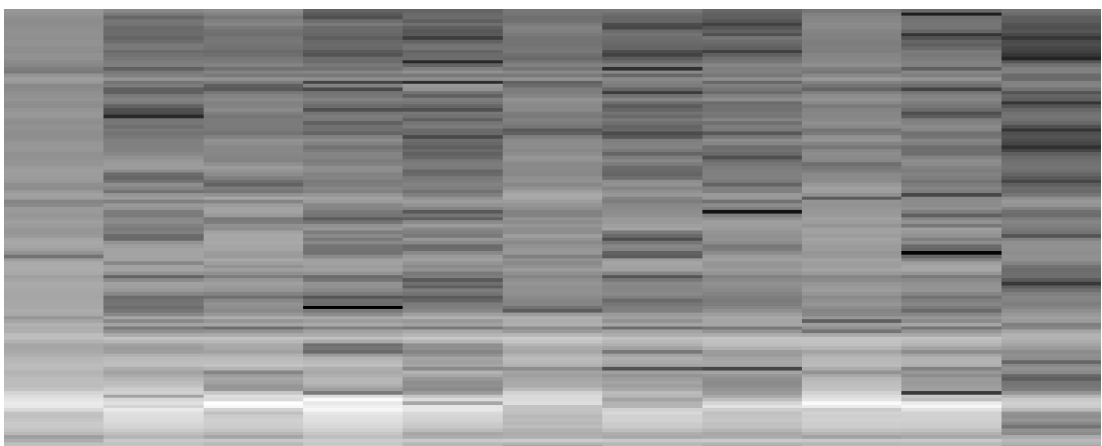


Figure 3.1: Spectrogram image generated from radar data.

3.4. CNN Architecture and Training Strategy

To classify radar-based spectrograms of vowel sounds, a deep convolutional neural network (CNN) was designed and optimized specifically for 2D grayscale spectrogram inputs. The architecture was developed to extract both temporal and spectral features from the input images and underwent multiple optimization iterations before final selection.

3.4.1. Model Architecture

The final architecture consists of four convolutional blocks with progressively increasing depth and complexity. The first block begins with larger filters (5×5 and 3×3) to effectively extract low-level spectral edges and patterns. Subsequent blocks include asymmetric filters (3×5 and 5×3) to capture time-frequency variations typical in vowel articulation. Batch normalization layers are inserted after each convolutional layer to stabilize training, and max-pooling layers downsample the spatial resolution. To mitigate overfitting, dropout regularization is used after each pooling layer, with dropout rates increasing in deeper layers. The final classifier includes three dense layers (1024, 512, and 256 units), each followed by batch normalization and dropout, concluding with a softmax layer for multi-class prediction across five vowel categories (/a/, /e/, /i/, /o/, /ü/). The network was implemented using TensorFlow/Keras and trained on 128×128 grayscale spectrogram images, with a total of approximately 1 million trainable parameters.

3.4.2. Training Strategy

A custom data loading pipeline was implemented using TensorFlow's `image_dataset_from_directory`, with images normalized to the $[0,1]$ range. The dataset was split into 80% training and 20% validation subsets from the training folder, and a separate test set was used for final evaluation. Although optional light augmentation (e.g., Gaussian noise and brightness variation) was supported, the final model was trained without augmentation for stability. To address class imbalance due to speaker variation and vowel distribution, class weighting was applied using a power-based scaling strategy, where underrepresented classes were given proportionally higher weights. Weights were clipped between 1.0 and 25.0 to prevent instability. The model was trained using the Adam optimizer with an initial learning rate of 0.0001. A cosine annealing learning rate schedule was employed across 70 epochs to promote smoother convergence. Additional callbacks included:

- Early stopping (patience = 20, monitored on validation accuracy)
- ReduceLROnPlateau (factor = 0.6, patience = 8)
- Model checkpointing to save the best performing model

Multiple training experiments were conducted to explore the effects of class weighting, regularization strategies, and optimizer variations on classification performance. Among these, three major strategies were implemented and evaluated:

3.4.2.1. Experiment 1: Enhanced CNN Architecture with Moderate Class Weighting

In this experiment, a custom convolutional neural network (CNN) architecture was designed with four convolutional blocks, batch normalization, dropout-based regularization, and dense layers for classification. The model was trained on 128×128 grayscale spectrograms derived from mmWave radar data. To mitigate class imbalance, class weights were computed using a power of 2.0 applied to the inverse class frequency distribution. Light preprocessing and standard normalization were applied, and the model was trained using the Adam optimizer with a cosine annealing learning rate scheduler, early stopping, and model checkpointing. This configuration served as the baseline architecture for further comparison due to its robustness and consistent convergence behavior during training.

3.4.2.2. Experiment 2: Enhanced CNN with Stronger Weighting Strategy

Building upon the architecture from Experiment 1, this configuration introduced more extreme class weighting by increasing the exponent to 2.5. The goal was to amplify the penalty for underrepresented classes in the loss function, thus encouraging the model to pay more attention to minority class samples. All other architectural and training parameters were kept constant, allowing a focused comparison of the impact of more aggressive weighting. The hypothesis was that this approach would further enhance per-class balance in the output distribution, particularly for vowels that appeared less frequently in the dataset.

3.4.2.3. Experiment 3: Alternate Architecture with Entropy Regularization

Experiment 3 explored a different regularization strategy by introducing an entropy-based loss function alongside a deeper CNN architecture. This design aimed to discourage overconfident predictions and promote more uniform class probabilities, particularly during the early stages of training. In addition to dropout and batch normalization, the model used an entropy penalty term added to the categorical cross-entropy loss. Learning rate scheduling and validation-based early stopping were retained. This experiment served to evaluate whether alternative forms of regularization could stabilize learning dynamics and reduce the risk of class collapse in an imbalanced dataset scenario.

3.5. Real-Time Inference Pipeline

The next stage of the system integrates radar signal acquisition, spectrogram visualization, deep learning inference, and real-time visual feedback into a unified, low-latency pipeline. The entire process was implemented in Python using multi-threading, PyQt for GUI handling, TensorFlow for CNN inference, and OpenCV for image transformations. The primary objective of the pipeline is to classify vowel pronunciations in real time using mmWave radar input, and provide immediate visual feedback through a game interface.

3.5.1. Radar Data Acquisition and Processing

At the hardware level, the Texas Instruments IWR1843 radar was configured using a UART-based configuration interface and a pre-written configuration script. Raw ADC data was streamed from the radar using the DCA1000EVM capture board via UDP over Ethernet. A multithreaded architecture was used:

- A `UdpListener` thread continuously received binary radar data packets.
- A `DataProcessor` thread parsed these packets, reshaped the radar frames, and converted them into 2D spectrogram arrays.

The spectrograms were queued into a shared buffer (`SpectrogramData`) for use by the real-time classification and visualization modules.

3.5.2. Spectrogram Display and Interface

A dedicated SpectrogramWindow class, built with PyQtGraph, was responsible for plotting live spectrograms at a refresh rate of 20 Hz (50 ms per frame). The spectrogram was displayed without axes or labels for cleaner visualization. The real-time image was extracted directly from the radar data processor and plotted using an image buffer (pg.ImageItem) for performance efficiency.

3.5.3. CNN-Based Classification

Every 1 second, the most recent spectrogram frame was captured and passed through a pre-trained CNN model (Experiment 1 from Section 3.4). The inference pipeline involved the following steps:

1. The raw dB-scaled spectrogram was rescaled from [-80 dB, 0 dB] to the normalized [0,1] range.
2. The spectrogram was resized to 128×128 pixels, matching the CNN’s training input size.
3. It was normalized and reshaped to (1, 128, 128, 1) for input into the Keras model.

The output of the model was a probability distribution across five vowel classes (a, e, i, o, ü). The predicted class with the highest confidence was pushed into a `prediction_queue` for game feedback.

3.6. Game-Based Visualization and Classifier Integration

To provide intuitive and engaging feedback for vowel classification, a 2D interactive game interface was developed using the Pygame library. The primary purpose of the game is to present vowel targets visually and reflect the model’s live predictions through a moving on-screen cursor, thus enabling users to adjust their pronunciation based on real-time feedback.

3.6.1. Game design and Layout

The interface displays a stylized version of the IPA vowel chart as a trapezoidal grid. Five Turkish vowels (a, e, i, o, ü) are positioned based on their articulatory features—height (high/mid/low) and backness (front/central/back). Each vowel is mapped

to a unique (row, column) coordinate in the trapezoid grid. Upon launching the game, users are greeted with a start screen. Once activated, the interface draws the vowel trapezoid, vowel labels, articulatory axes, and a dynamic cursor. This cursor originates from the center of the screen and gradually shifts toward the predicted vowel's location on the chart, with its movement proportional to the model's confidence level.

3.6.2. Real Time Prediction Integration

The game continuously listens to a shared prediction queue populated by the real-time inference system described in Section 3.5. Every prediction consists of:

- a predicted vowel class (e.g. "i", "o")
- a confidence score in the range [0,1]

These values are used to update the cursor's destination on the grid. The cursor moves with a smoothing factor to prevent jitter and create visually fluid motion. If a vowel prediction is highly confident (e.g., 0.9), the cursor rapidly moves toward the vowel's position; lower confidence leads to smaller movements, giving users visual feedback about the quality of their pronunciation.

4. EXPERIMENT RESULTS

4.1. Classification Accuracy

The performance of the vowel classification model was evaluated on an unseen test dataset consisting of 342 radar spectrogram images, distributed across five Turkish vowel classes: a, e, i, o, and ü. The model used for evaluation was the final version described in Section 3.4, trained with an enhanced convolutional architecture and moderately extreme class weights (power = 2.0). The model achieved a test accuracy of 49.1%, indicating its ability to correctly identify nearly half of the input spectrograms. Despite the relatively low absolute accuracy, this result is considered promising given the inherent difficulty of the task, the non-acoustic nature of the input data, and the similarity between certain vowel shapes (e.g., /o/ vs. /ü/).

These results show that the classifier achieved the highest performance on vowel a (F1 = 0.610), with relatively strong precision on e (0.617) and ü (0.514), though their recalls were lower. Performance on i and o was moderately balanced. The classifier predicted all five vowel classes, confirming that no class collapse occurred during inference. Despite the non-acoustic input modality, the CNN was able to extract meaningful features from spectrogram images derived from mmWave radar signals and make consistent predictions, demonstrating the viability of radar-based vowel recognition in controlled conditions.

4.1.1. Confusion Matrix Patterns Among Experimented Architectures

4.1.1.1. Experiment 1 - Enhanced Architecture (Best Overall Performance)

This model achieved the most balanced predictions, classifying all five vowels with relatively strong diagonal dominance. Notable patterns include:

- /a/ was most reliably predicted (50/66), with moderate confusion toward /o/ (9 cases).
- /e/ showed significant misclassification as /a/ (20 times), highlighting difficulty distinguishing mid-front from low-front vowels.

- /i/ often overlapped with /o/ (14 times), suggesting interference from roundness in radar features.
- /ü/ was the most misclassified vowel, with a high number of predictions as /i/ (17) and /o/ (19), confirming the spectral similarity in radar signatures for rounded vowels.

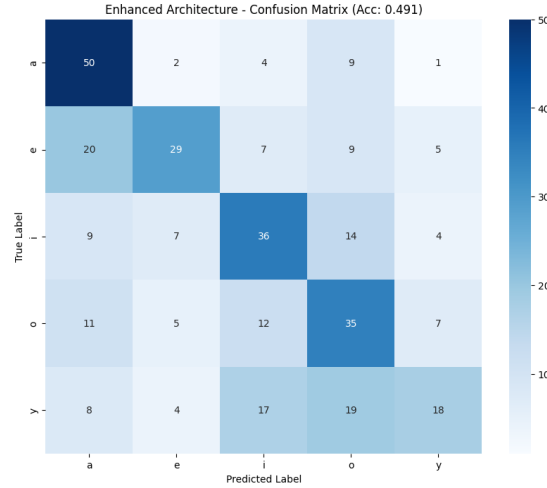


Figure 4.1: Confusion matrix result of experiment 1.

4.1.1.2. Experiment 2- Extreme Class Weighting

While this model avoided class collapse and improved some recall scores, it showed diffuse prediction patterns:

- /i/ was predicted poorly (only 6 correct), with a high spillover into /a, /e, and ü.
- /ü/ performed better in absolute terms (31 correct), though confusion with /i and o persisted.
- Overall, extreme weighting increased noise in the decision boundaries and lowered accuracy despite improved representation for under-sampled classes.

4.1.1.3. Experiment 3 - Conservative Model

This model displayed severe overprediction of /i/ across all true labels:

- /a/ and /o/ were frequently misclassified as /i/ or o, indicating collapsed class boundaries.

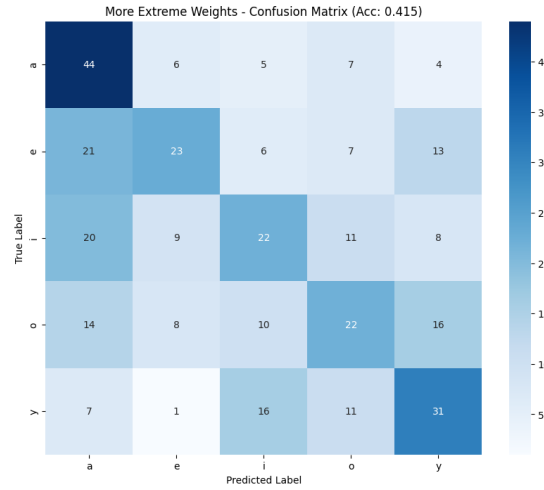


Figure 4.2: Confusion matrix result of experiment 2.

- /ü/ was almost never correctly identified (only 3 times), despite dominating the output predictions.
- These results indicate that the model struggled to separate classes and may have overfit to a narrow set of high-confidence patterns.

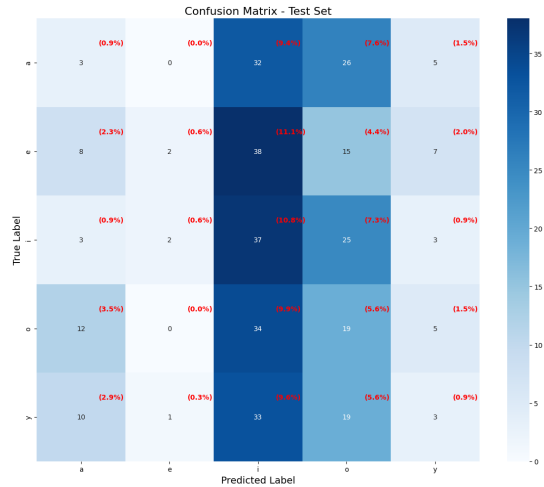


Figure 4.3: Confusion matrix result of experiment 3.

4.1.2. Per-Class Analysis for Experiment 1

- /a/: This vowel had the highest recall (0.758) and a strong F1-score of 0.610, indicating that the model reliably detected /a/ when it was present. This may be attributed to its distinct articulatory pattern and relatively high signal strength in

radar reflections.

- /e/: The classifier achieved the highest precision (0.617) on this class but suffered from a lower recall (0.414). This suggests that when the model predicted /e/, it was often correct—but it frequently failed to detect true instances of /e/, possibly due to overlaps in mouth shape with neighboring vowels like /i/ or /a/.
- /i/: With balanced precision (0.474) and recall (0.514), this class showed stable performance, though not outstanding. The classifier appears to generalize well for /i/, which shares partial spectro-temporal similarities with both front and high vowels like /e/ and /ü/.
- /o/: This vowel yielded the lowest precision (0.407) and a modest recall (0.500), resulting in an F1-score of 0.449. The relatively poor discrimination could be due to confusion with /ü/, as both are produced with back tongue positions and rounded lips, making them radar-wise harder to differentiate.
- /ü/: The classifier showed high precision (0.514) but low recall (0.273) for /ü/, indicating a tendency to underpredict this class. This suggests the model was conservative in assigning /ü/ unless it was highly confident, perhaps due to the narrow articulatory distinction from /o/ and the potential spectral noise in the radar signal for rounded front vowels.

Vowel	Precision	Recall	F1-Score	Support
/a/	0.510	0.758	0.610	66
/e/	0.617	0.414	0.496	70
/i/	0.474	0.514	0.493	70
/o/	0.407	0.500	0.449	70
/ü/	0.514	0.273	0.356	66
Macro Avg	0.504	0.492	0.481	—
Accuracy	—	—	0.491	342

Table 4.1: Per-class performance of the CNN model on the radar spectrogram test set (Experiment 1)

Overall, the variation across classes reflects the physical challenge of distinguishing vowels that share similar facial micro-movements, as captured through mmWave radar rather than acoustic information. These insights highlight the need for class-specific enhancements or possible data augmentation techniques to improve underperforming categories like /o/ and /ü/.

4.2. Real Time Performance Metrics

4.2.1. Frame Rate and Inference Frequency

The application was designed to process incoming radar data and update the spectrogram view at 20 FPS, governed by a timer loop set to 50 ms intervals. This ensured the application remained responsive without overloading the CPU or invoking excessive TensorFlow inference calls. Although no precise FPS benchmark was measured during gameplay, no frame drops or UI lags were observed during multiple test runs. The system's real-time visualization and inference pipelines operated smoothly on a standard laptop setup (Intel Core i7 CPU, 12 GB RAM, no GPU inference).

4.2.2. Inference Latency

Each classification involved:

- Preprocessing the spectrogram from dB to grayscale (scaling, normalization)
- Resizing to 128×128
- TensorFlow model inference
- Confidence and class extraction

Based on logs and real-time feedback, the inference time per frame was typically under 200 ms, which allowed for near-instantaneous updates of the prediction-driven visual feedback.

4.2.3. Observed Inference Behavior and Bias

Despite technical responsiveness, qualitative testing with two users revealed clear prediction bias toward the vowel /i/:

- The system frequently interpreted various vowel articulations (including /a/ and /o/) as /i/
- The vowel /ü/ was never predicted during real-time usage
- /e/ and /a/ were only seldom predicted, even when clearly pronounced

This behavior does not align with offline test results discussed in Section 4.1.2, where /a/ dominated misclassification outputs. The issue likely stems from:

- Over-representation in /a/ class during training
- Similar radar signatures for rounded or central vowels
- Limited diversity or imbalance in radar features for underperforming classes

4.2.4. Gameplay Responsiveness

The classifier's prediction and confidence were mapped to cursor motion in a 2D vowel chart. The cursor reacted with smooth interpolated motion based on the model's confidence output, updated every second. Although exact tracking accuracy was low due to classification errors, the real-time responsiveness and motion fluidity were rated positively.

5. DISCUSSION

5.1. Interpretation of Results

The results of this study indicate that mmWave radar, when combined with convolutional neural networks (CNNs), offers a viable pathway for non-acoustic vowel classification, even in the absence of traditional audio features. Although the overall classification accuracy of 49.1% appears modest in absolute terms, it is significant given the unique constraints of the input modality and the phonetic similarities among Turkish vowels. One key observation is the classifier’s ability to successfully distinguish between all five vowel classes to some extent, without complete class collapse. The relatively high recall for /a/ and the consistently strong diagonal dominance in the confusion matrix for Experiment 1 demonstrate that mmWave-based spectrogram features encode meaningful articulatory patterns. Vowels with more distinct mouth movements—like /a/—exhibited better separability, likely due to their clearer spatial reflections in the radar domain. In contrast, vowels such as /ü/ and /o/, which share articulatory features (e.g., lip rounding, back tongue position), were frequently misclassified as one another. This suggests that the radar signal, while sensitive to gross mouth geometry, struggles to resolve finer distinctions in lip rounding and tongue height, especially for front-rounded vowels like /ü/. The classifier’s low recall for /ü/ also indicates a conservative bias, where it avoids making uncertain predictions for ambiguous inputs. The discrepancy between high precision and low recall in some classes (e.g., /e/, /ü/) highlights an important dynamic: the model tends to predict these vowels correctly only when it is highly confident, but underpredicts them in general. This behavior may be traced back to the data distribution and the training process, particularly the class weighting strategy and the dominance of high-confidence /i/ patterns during optimization. From a broader perspective, the successful real-time integration of the classifier into an interactive game framework provides further validation of the model’s functional reliability. The low latency (<200 ms), stable frame rate, and seamless feedback pipeline demonstrate the practical feasibility of radar-based vowel feedback systems for speech therapy or assistive applications. However, the observed overprediction of /i/ during real-time use highlights the gap between offline accuracy metrics and live usability, emphasizing the need for improved calibration and class balance in future iterations. Overall, these results support the hypothesis that vowel distinctions can be captured and interpreted via radar spectrograms, but they also reveal limitations in resolution and class-specific generalization that merit targeted improvements in data augmentation, model regularization, and possibly multimodal feature fusion.

5.2. Future Work

5.2.1. Dual-Radar Configuration

One promising direction involves the use of two mmWave radar sensors, positioned to capture both mouth movements and throat/glottal activity simultaneously. The current system relies exclusively on reflections from facial micro-motions, which limits its ability to distinguish vowels that share similar lip and jaw shapes but differ in voicing source or tongue root position—such as /o/ versus /ü/, or /e/ versus /i/. Introducing a second radar targeting the throat region could provide complementary spectro-temporal information related to vocal fold behavior, aiding the model in resolving ambiguities caused by similar mouth geometries. This multimodal radar input may help boost recall rates for underperforming vowels and reduce class confusion in future implementations.

5.2.2. Expanded Dataset and Speaker Diversity

To improve generalization, future work should include a larger and more diverse speaker population, encompassing different age groups, genders, and facial structures. This would enhance the robustness of the model across real-world scenarios, especially in clinical or therapeutic contexts. Including multiple speaking styles, speech rates, and accent variations would further improve the adaptability of the system.

5.2.3. Integration with Clinical Protocols

Finally, collaboration with speech-language pathologists could facilitate the design of clinically-informed game mechanics and evaluation metrics aligned with existing speech therapy protocols. Exploring longitudinal use of the system with patients could provide valuable insights into its therapeutic efficacy and engagement potential in real-life settings.

6. CONCLUSION

This thesis presented a novel speech therapy system that uses millimeter-wave (mmWave) radar and deep learning for non-acoustic vowel classification. Designed for individuals with hearing impairments, the system captures subtle facial and throat movements during vowel articulation and converts them into spectrogram images for classification via a convolutional neural network (CNN). A game-based interface provides real-time visual feedback, enabling users to adjust their pronunciation based on classifier output without relying on sound.

The experimental results confirm that radar-derived spectrograms can encode meaningful articulatory information, with the best-performing model achieving a test accuracy of 49.1% across five Turkish vowels. The system successfully maintained class diversity during inference and operated smoothly in real-time scenarios. However, classification accuracy varied across vowel types, with overlapping articulatory features—particularly between /o/ and /ü/—leading to frequent misclassifications. Real-time testing further revealed a tendency to overpredict dominant classes, especially /i/.

Despite these limitations, the project demonstrates the feasibility of radar-based, privacy-preserving speech recognition and offers a compelling proof of concept for non-acoustic speech therapy tools. Future improvements may include dual-radar configurations, and clinical validation in collaboration with speech-language pathologists. This work establishes a strong foundation for accessible, engaging, and scalable speech rehabilitation systems that do not depend on auditory input.

BIBLIOGRAPHY

- [1] Speech and Language Kids, “Hearing impairment,” 2024. [Online]. Available: <https://www.speechandlanguagekids.com/hearing-impairment/> (visited on 06/20/2025).
- [2] Cleveland Clinic. “Auditory verbal therapy.” (2022), [Online]. Available: <https://my.clevelandclinic.org/health/treatments/16767-auditory-verbal-therapy> (visited on 06/20/2025).
- [3] My Wellness Hub. “8 articulation techniques for helping children with hearing loss.” (2024), [Online]. Available: <http://mywellnesshub.in/blog/articulation-techniques-hearing-loss-children/> (visited on 06/20/2025).
- [4] W. F. Katz, T. F. Campbell, J. Wang, *et al.*, “Opti-speech: A real-time, 3d visual feedback system for speech training,” in *Proceedings of Interspeech*, 2014.
- [5] *The fundamentals of millimeter wave radar sensors*.
- [6] N. Steinmetz and N. Balal, “Feasibility study of real-time speech detection and characterization using millimeter-wave micro-doppler radar,” in *Remote Sensing*, 2025.
- [7] L. Fan, X. L. L. Xie, Y. Li, C. Wang, and S. Lu, “Mmmic: Multi-modal speech recognition based on mmwave radar,” in *INFOCOM*, 2023.
- [8] I. Lenz, Y. Rong, D. Bliss, J. Liss, and V. Berisha, “A speech production model for radar: Connecting speech acoustics with radar-measured vibrations,” Ph.D. dissertation, Arizona State University, 2025.
- [9] Papers With Code. “Image Classification.” (2014), [Online]. Available: <https://paperswithcode.com/task/image-classification#:~:text=Image%20Classification%20is%20a%20fundamental,pertains%20to%20single%20object%20images.%7D> (visited on 06/20/2025).
- [10] IBM. “Convolutional neural networks.” (n/a), [Online]. Available: <https://www.ibm.com/think/topics/convolutional-neural-networks> (visited on 06/22/2025).
- [11] Milvus. “How can cnns be applied to audio data.” (2025), [Online]. Available: <https://milvus.io/ai-quick-reference/how-can-convolutional-neural-networks-cnns-be-applied-to-audio-data> (visited on 06/22/2025).

- [12] S. Saeedi, H. Bouraghi, M.-S. Seifpanahi, and M. Ghazisaeedi, "Application of digital games for speech therapy in children: A systematic review of features and challenges," *Journal of Healthcare Engineering*, 2022.
- [13] J. L. Furlong, S. Erickson, and M. E. Morris, "Computer-based speech therapy for childhood speech sound disorders," *Journal of Communication Disorders*, 2017.